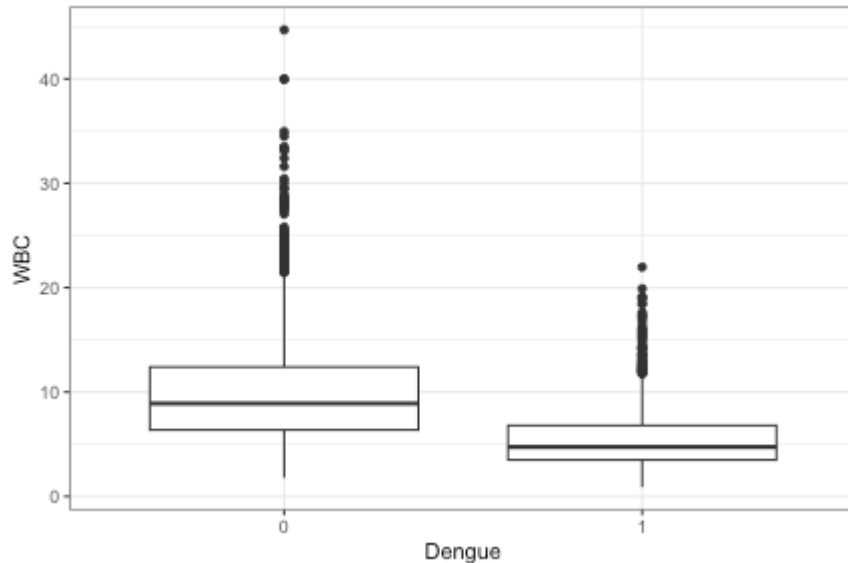


Inference

Motivation



It looks like there is a difference in WBC between positive and negative dengue cases.

How would we assess whether this is real, or due to chance?

Motivation

Let μ_0 be population mean WBC for non-dengue cases, and μ_1 be population mean WBC for dengue cases.

Want: $\mu_1 - \mu_0$

Estimate:

$$\overline{WBC}_1 - \overline{WBC}_0 = -4.36$$

How do we account for variability between samples??

Confidence interval

A **confidence interval** gives a "reasonable range" of values for an unknown parameter.

Want: $\mu_1 - \mu_0$

Estimate:

$$\overline{WBC}_1 - \overline{WBC}_0 = -4.36$$

Confidence interval: estimate $\pm$ critical value $\times$ SE

need $SE(\overline{WBC}_1 - \overline{WBC}_0)$

Ideally: Take many samples from population;
calculate $\overline{WBC}_1 - \overline{WBC}_0$ for each sample;
look at variability between samples

Problem: only get one sample from population!

Bootstrap resampling

- Resample rows of the original data
(with replacement)
- Compute quantity of interest (e.g., $\overline{WBC}_1 - \overline{WBC}_0$)
on bootstrap sample
- Repeat many times
- estimate of SE is std dev. of bootstrap
statistics

Variance of statistic between samples from population $\approx$ Variance of statistic between bootstrap samples

Bootstrap resampling

A small subset of the original data:

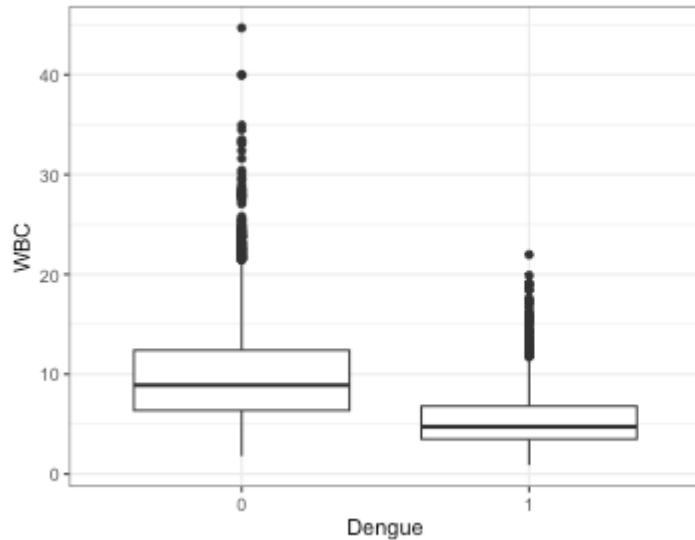
##		WBC	PLT	Age	Dengue
##	1	5.10	183	3	0
##	2	10.20	279	5	0
##	3	6.48	195	5	0
##	4	7.54	269	3	0
##	5	3.60	191	5	1

A bootstramp sample of these rows:

##		WBC	PLT	Age	Dengue
##	1	5.10	183	3	0
##	2	6.48	195	5	0
##	3	6.48	195	5	0
##	4	3.60	191	5	1
##	5	5.10	183	3	0

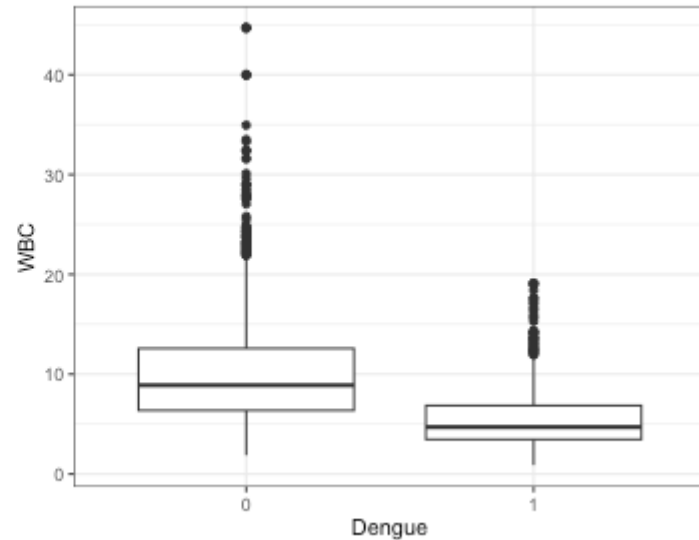
Bootstrap resampling

original data



$$\overline{WBC}_1 - \overline{WBC}_0 = -4.36$$

bootstrap sample

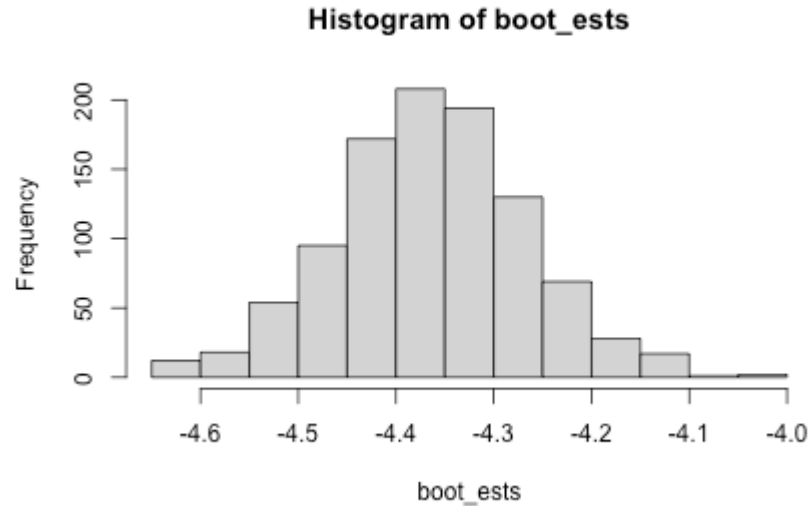


$$\overline{WBC}_1 - \overline{WBC}_0 = -4.47$$

Activity

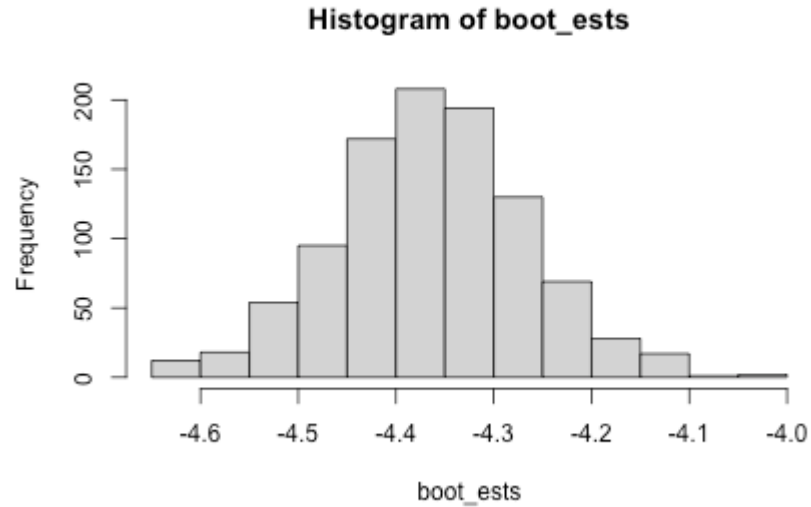
- + Work on the class activity (linked on Canvas and course website)
- + Submit rendered HTML on Canvas at the end of class

Activity



Based on the bootstrap samples, what would be a reasonable range of values for the difference in means?

Activity



What does the distribution look like?

$\approx$ Normal!

Activity

Bootstrap percentile interval:

```
quantile(boot_ests, c(0.025, 0.975))
```

```
##          2.5%          97.5%  
## -4.559510 -4.172119
```

Normal interval with bootstrap SE:

```
-4.36 - 1.96*sd(boot_ests)
```

```
## [1] -4.551451
```

```
-4.36 + 1.96*sd(boot_ests)
```

```
## [1] -4.168549
```

Activity

Comparison with the usual formula:

$$\widehat{SE}(\bar{X}_1 - \bar{X}_0) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}$$

```
sqrt(var(dengue$WBC[dengue$Dengue == 1])/sum(dengue$Dengue == 1) +  
      var(dengue$WBC[dengue$Dengue == 0])/sum(dengue$Dengue == 0))
```

```
## [1] 0.1034442
```

Bootstrap estimate of standard error:

```
sd(boot_ests)
```

```
## [1] 0.09767924
```

```
]
```